Edit

Data Science

Data Science & Machine Learning
Turning Materials Data into Predictive Insights

Data science is the detective work of the digital age — extracting actionable knowledge from complex, noisy datasets to solve real-world problems and accelerate discovery.

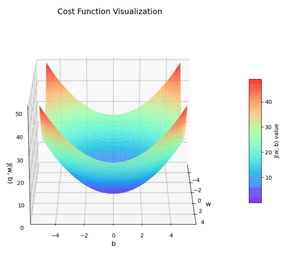
In my research, it serves as a powerful bridge between traditional mechanical/materials engineering and modern computation:

- Predicting material properties and failure modes without exhaustive physical experiments
- Optimizing additive manufacturing parameters (e.g., LPBF/DED settings for NiTi or TPMS lattices)
- Analyzing large experimental datasets from mechanical testing, microscopy, and thermal analysis
- Building models for shape memory behavior, metamaterial performance, or process reproducibility

By combining statistics, programming, and machine learning, I aim to make materials research faster, more efficient, and more innovative — especially in additive manufacturing and smart structures.

Key Projects & Capstones

1. **Supervised Machine Learning: Regression and Classification from Scratch**
   GitHub: https://github.com/drshahadathussain/supervised_machine_learning_regression_and_classification



   Built foundational ML models entirely from scratch (no high-level libraries for core algorithms) to deepen understanding of supervised learning.
   - Regression: Modeled continuous relationships (e.g., predicting house prices or material strength) using linear/polynomial regression, gradient descent, vectorization, and regularization (L2 to combat overfitting).
   - Classification: Handled binary decisions (e.g., defect detection or success/failure) with logistic regression.
   - Outcomes: Achieved low MSE in regression via feature engineering/polynomials; 85–95% accuracy in classification tasks with regularization reducing overfitting.
   - Tools: Python, NumPy (for linear algebra), scikit-learn (validation/comparison), Jupyter Notebooks.

   This work forms the bedrock for applying ML to materials datasets — e.g., predicting lattice mechanical response from design parameters.
2. **Falcon 9 First-Stage Booster Landing Success Prediction**
   GitHub: https://github.com/drshahadathussain/capstone-applied-data-science
   End-to-end data science project predicting whether SpaceX Falcon 9 boosters land successfully (binary classification: 1 = success, 0 = failure). Used real historical launch data to support cost/reusability analysis in aerospace.

   Source: Google images

   - Data pipeline: Collected via SpaceX API + web scraping; cleaned, encoded, and standardized features (payload mass, orbit type, launch site, booster version, etc.).
   - EDA & Visualization: Statistical summaries, Folium maps for launch sites, Plotly Dash interactive dashboards.
   - Modeling: Trained Logistic Regression, SVM, Decision Trees, KNN; used cross-validation, hyperparameter tuning, and confusion matrices.
   - Key Insights: 80–95% accuracy (typically 83–90%); higher success rates with 2000–4000 kg payloads, KSC LC-39A site, equatorial orbits, and post-2015 boosters.
   - Tools: Python, Pandas, NumPy, Matplotlib/Seaborn, Folium, Plotly Dash, scikit-learn, Requests/BeautifulSoup, SQLite.

   Demonstrates full DS workflow — from raw data to business insights — transferable to predicting AM build success, material fatigue, or lattice performance.

Core Data Science Skills & Tools

- Languages & Environments: Python, Jupyter Notebooks, Linux/Shell Scripting
- Data Handling: Pandas, NumPy, SciPy, SQLite, data wrangling/cleaning, feature engineering
- Visualization: Matplotlib, Seaborn, Plotly, Folium (geospatial), Dash (interactive apps)
- Machine Learning: Supervised learning (regression/classification), scikit-learn, model evaluation/tuning, regularization, gradient descent, overfitting mitigation
- Other: API/web scraping (Requests, BeautifulSoup), SQL querying, geospatial analysis

Certifications & Learning Path

- IBM Data Science Professional Certificate (Coursera)
- University of Michigan – Python for Everybody Specialization (Coursera)
- Google AI Essentials Certificate

Why This Matters for My Research
Data science is no longer optional in materials engineering — it's essential. Whether modeling the phase transformation in NiTi TPMS lattices, predicting flexural behavior of gyroid structures, or optimizing DED parameters from sensor data, ML and statistical tools let us move from trial-and-error to predictive, data-informed design.

I'm actively expanding this focus: applying ML to experimental datasets from mechanical testing, microstructural analysis, and additive processes. Future updates here will include new notebooks, materials-specific models, and insights from ongoing work.

Subscribe for alerts on new data science posts, code releases, or applications to metamaterials/AM.

Explore full portfolio → www.shahadathussain.com/data-science
GitHub: drshahadathussain

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